Provenance-Aware Agentic Knowledge Graph Systems for Reliable Multi-Source Grounded AI

Primary Topic: Knowledge Grounding **Secondary Topic:** Agentic AI

Retrieval-augmented generation (RAG) [1] has become a foundational technique for grounding large language model (LLM) inference in external knowledge. Production systems have matured substantially: hybrid retrieval combining semantic (dense) and keyword (sparse) search is now standard practice [2], and retrieval is increasingly exposed as a tool within agentic workflows where models decide *when* and *what* to retrieve [3]. Yet despite these advances, two critical gaps remain. First, even state-of-the-art retrieval systems lack structured reasoning substrates capable of multi-hop synthesis across heterogeneous knowledge sources—a gap that graph-based approaches like GraphRAG [4] begin to address but do not fully resolve. Second, current systems provide limited provenance: they can cite documents, but cannot produce fine-grained, verifiable evidence chains linking individual model outputs to specific supporting knowledge, and they lack validation mechanisms that detect and recover from retrieval failures [5], [6].

This project proposes **Provenance-Aware Agentic Knowledge Graph Systems (PA-AKG)**, an architecture that unifies structured knowledge graph (KG) reasoning, adaptive multi-source retrieval, and agent-driven validation into a single grounding substrate—capabilities that existing systems provide only in isolation [4], [5], [7]. The core contribution is a framework in which agents reason over entities, relationships, and structured constraints while maintaining explicit, verifiable evidence chains from outputs to specific supporting knowledge. Agents dynamically select among retrieval pathways—dense, sparse, structured graph queries, and hybrid combinations—based on task context and confidence, enabling reasoning workflows that balance semantic recall with symbolic precision.

The proposed system enables **multi-source knowledge grounding** across knowledge graphs, relational databases, scientific literature, and domain-specific document collections. A critical component is an automated **provenance generation and validation layer** that produces fine-grained evidence links from model outputs to specific supporting knowledge sources—going beyond document-level citation toward span- and entity-level traceability. Provenance signals are structured to align with emerging observability standards [8], making agent reasoning auditable across tool calls and retrieval actions. Validation agents monitor retrieval quality and trigger corrective actions when evidence is weak, contradictory, or insufficient, extending retrieval-control mechanisms [5], [7] from unstructured document retrieval to structured knowledge graph reasoning.

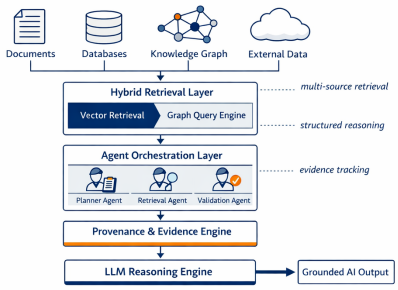

Fig. 1. **PA-AKG Architecture.** Heterogeneous knowledge sources feed a hybrid retrieval layer (vector search + KG queries). An agent orchestration layer coordinates planning, retrieval, and validation agents. A provenance engine records evidence links aligned with observability standards [8] for traceable verification.

This research investigates: (1) how heterogeneous knowledge sources can be unified into a grounding framework supporting both semantic and structured retrieval; (2) how agent workflows should route and validate across retrieval pathways under uncertainty; (3) what mechanisms enable scalable fine-grained provenance generation tied to specific knowledge entities; and (4) whether provenance-aware KG grounding yields measurable gains in reasoning accuracy, output verifiability, and retrieval failure recovery relative to hybrid RAG baselines.

We will implement the PA-AKG architecture with integrated validation and provenance enforcement, and develop benchmarks measuring grounded reasoning quality, evidence completeness, and retrieval failure recovery. Experiments compare vector-based RAG, hybrid retrieval, GraphRAG-style reasoning, and PA-AKG across **multi-hop QA over heterogeneous knowledge bases, claim verification with evidence chain reconstruction, and engineering decision-support** tasks. Expected outcomes include the PA-AKG open prototype, provenance-aware reasoning benchmarks, and baselines advancing **reliable, transparent, and auditable agentic AI**.

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